

1. (80 points) Using Tables 3-1 and 3-2, or by direct calculation, determine the following:

- a. The Laplace transform of $(t-2)u(t)$
- b. The inverse Laplace transform of $\frac{e^{-s}}{s^2}$
- c. The Laplace transform of $(t-2)e^{-t}u(t)$
- d. The Laplace transform of $(t-1)e^{-(t-1)}u(t-1)$
- e. The inverse Laplace transform of $\frac{1}{s^2+16}$
- f. The Laplace transform of $e^{-2t} \cos(3t+135^\circ)u(t)$
- g. The Laplace transform of $\frac{d}{dt}\{t^2 e^{-t}u(t)\}$
- h. The Laplace transform of $\frac{d^3 x(t)}{dt^3}$, for a causal signal $x(t)$
- i. The Laplace transform of $r(t-2)$, where $r(t)$ represents the unit ramp signal
- j. The Laplace transform of $(t-1)^2 e^{-2(t-1)} \cos(5(t-1))u(t-1)$

2. (10 points) Calculate the initial and final values for the system $H(s) = \frac{10s+3}{s^2+3s+4}$

3. (10 points) Calculate the initial and final values for the system $H(s) = \frac{s+4}{s^2}$

1. (80 points) Using Tables 3-1 and 3-2, or by direct calculation, determine the following:

a. The Laplace transform of $(t-2)u(t)$

$$(t-2)u(t) = t u(t) - 2u(t) = \frac{1}{s^2} - \frac{2}{s}$$

b. The inverse Laplace transform of $\frac{e^{-s}}{s^2}$

$$\frac{e^{-s}}{s^2} = (t-1)u(t-1)$$

c. The Laplace transform of $(t-2)e^{-t}u(t)$

$$(t-2)e^{-t}u(t) = t e^{-t}u(t) - 2e^{-t}u(t) = \frac{1}{(s+1)^2} - \frac{2}{(s+1)}$$

d. The Laplace transform of $(t-1)e^{-(t-1)}u(t-1)$

$$(t-1)e^{-(t-1)}u(t-1) \Rightarrow e^{-t} \cdot t u(t) \Rightarrow \frac{e^{-s}}{(s+1)^2}$$

e. The inverse Laplace transform of $\frac{1}{s^2+16}$

$$\frac{1}{s^2+16} \Rightarrow \frac{1}{4} \left(\frac{1}{s^2+4^2} \right) = \frac{1}{4} \left(\frac{1}{s^2+4^2} \right) \Rightarrow \frac{1}{4} \sin(4t)u(t)$$

f. The Laplace transform of $e^{-2t} \cos(3t+135^\circ)u(t)$

$$e^{-2t} \cos(3t+135^\circ)u(t) = \frac{(s+2) \cos(-135^\circ) + 3s \sin(-135^\circ)}{(s+2)^2 + 9} = \frac{-(s+2) \cdot \frac{\sqrt{2}}{2} - 3 \cdot \frac{\sqrt{2}}{2}}{(s+2)^2 + 9}$$

g. The Laplace transform of $\frac{d}{dt} \{t^2 e^{-t}u(t)\}$

$$\frac{d}{dt} \{t^2 e^{-t}u(t)\} = s \left(\frac{2}{(s+1)^3} \right) - 0 = s \left(\frac{2}{(s+1)^3} \right)$$

h. The Laplace transform of $\frac{d^3 x(t)}{dt^3}$, for a causal signal $x(t)$

$$\frac{d^3 x(t)}{dt^3} = s^3 X(s) - s^2 x(0^-) - s x'(0^-) - x''(0^-)$$

Worksheet 7 cont.

i. The Laplace transform of $r(t-2)$, where $r(t)$ represents the unit ramp signal

$$r(t-2) = (t-2)u(t-2) = \frac{e^{-2s}}{s^2}$$

j. The Laplace transform of $(t-1)^2 e^{-2(t-1)} \cos(5(t-1))u(t-1)$

$$\begin{aligned} (t-2)^2 e^{-2(t-1)} \cos(5(t-1))u(t-1) &\Rightarrow t^2 e^{-2t} \cos(5t)u(t) \Rightarrow \frac{s+2}{(s+2)^2 + 25} \\ &\Rightarrow x''(s) \cdot e^{-5} \Rightarrow \frac{d^2}{ds^2} \left(\frac{s+2}{(s+2)^2 + 25} \right) \cdot e^{-5} \end{aligned}$$

2. (10 points) Calculate the initial and final values for the system $H(s) = \frac{10s+3}{s^2+3s+4}$

$$\text{Initial Value: } \lim_{s \rightarrow \infty} \left\{ \frac{10s^2+3s}{s^2+3s+4} \right\} = 10$$

$$\text{Final Value: } \lim_{s \rightarrow 0} \left\{ \frac{10s^2+3s}{s^2+3s+4} \right\} = 0$$

3. (10 points) Calculate the initial and final values for the system $H(s) = \frac{s+4}{s^2}$

$$\text{Initial Value: } \lim_{s \rightarrow \infty} \left\{ \frac{s^2+4s}{s^2} \right\} = 1$$

$$\text{Final Value: } \lim_{s \rightarrow 0} \left\{ \frac{s^2+4s}{s^2} \right\} = \infty$$